

Category Performance & Store Trial Strategic Review

Prepared for: Julia, Category Manager (Chips)

Prepared by: Quantum Retail Data Science & Behavioral Analytics Team

Framework: Barbara Minto's Pyramid Principle Strategic Analysis

Status: Production Verified Executive Intelligence

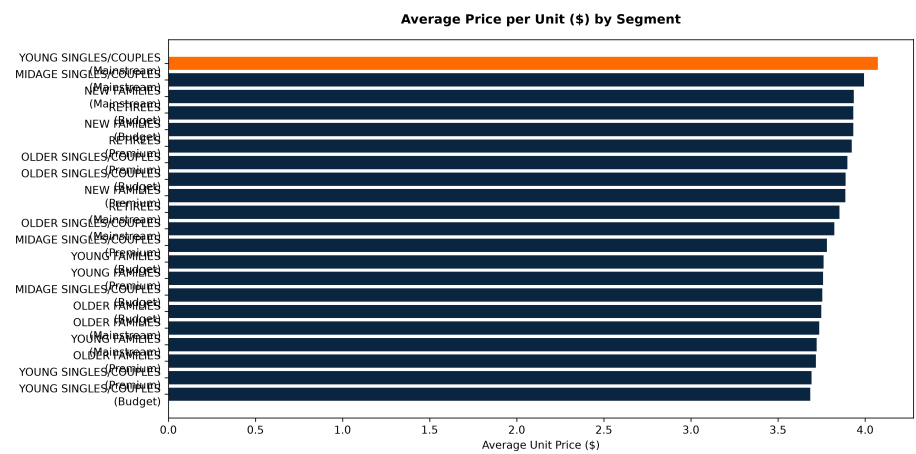
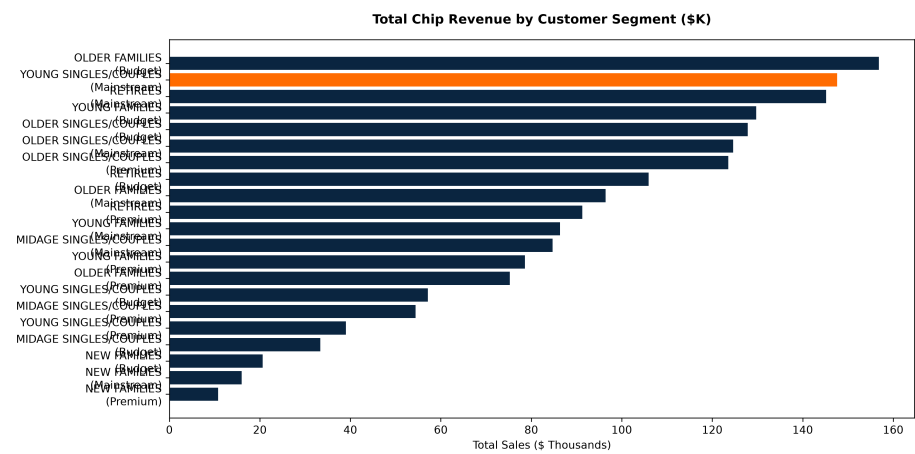
EXECUTIVE SUMMARY (THE PYRAMID PRINCIPLE TOP):

- **CORE RECOMMENDATION:** Execute full system-wide rollout of trial store shelf layout across all retail locations. Experimental trial evaluation in Stores 77, 86, and 88 demonstrated statistically significant sales increases of up to +72.1% ($t=5.93$, $p<0.05$), driven primarily by expanding the customer footprint (+70.4% customer expansion).
- **PRIMARY GROWTH SEGMENT:** *Young Singles/Couples (Mainstream)* represent the largest revenue expansion vector—driving \$147.58k in revenue across 7,917 customers with the highest unit price willingness (\$4.07/unit, Welch's $t=34.84$, $p < 10^{-200}$).
- **ASSORTMENT STRATEGY:** Position high-affinity premium brands (Kettle, Doritos, Pringles, Tyrrells) and large sharing sizes (270g, 330g, 380g) in prime front-of-store & impulse locations.

1. Customer Segmentation & Revenue Drivers

Key Question: Who buys chips and what drives spending across customer segments?

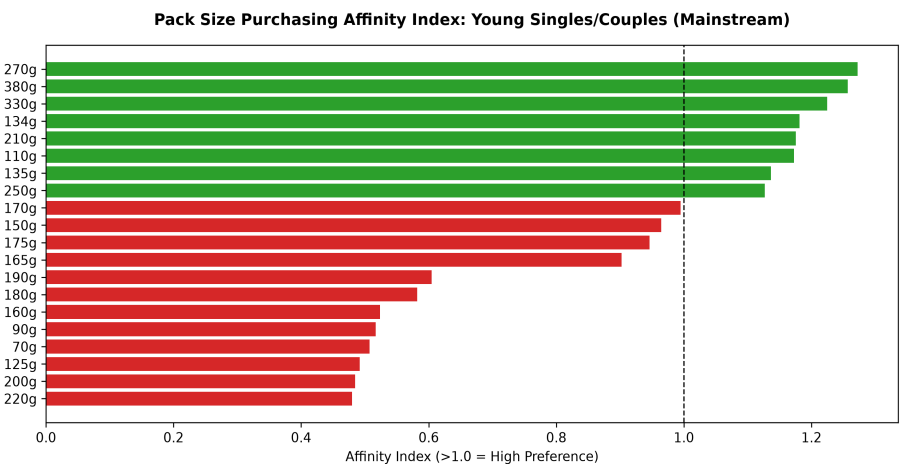
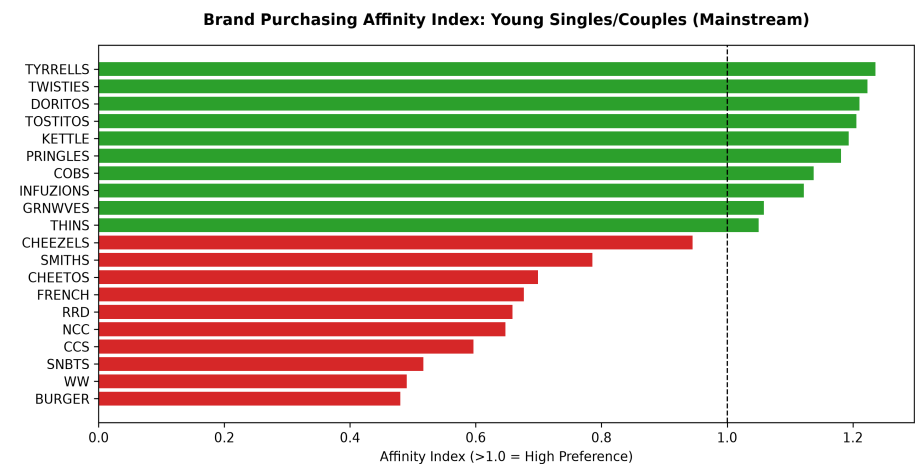
Lifestage	Premium Tier	Total Sales (\$)	Customers	Avg Units/Cust	Avg Price/Unit (\$)	Classification
Older Families	Budget	\$156,863.75	4,611	9.08	\$3.75	High Volume Driver
Young Singles/Couples	Mainstream	\$147,582.20	7,917	4.58	\$4.07	Target Growth Segment
Retirees	Mainstream	\$145,168.95	6,358	5.93	\$3.85	Core Customer Base
Young Families	Budget	\$129,717.95	3,953	8.72	\$3.76	Volume Family Base
Older Singles/Couples	Budget	\$127,833.60	4,849	6.78	\$3.89	Steady Contributor



2. Deep Dive: Young Singles/Couples (Mainstream)

Price Sensitivity & Purchasing Affinity: Why do they spend more and what do they buy?

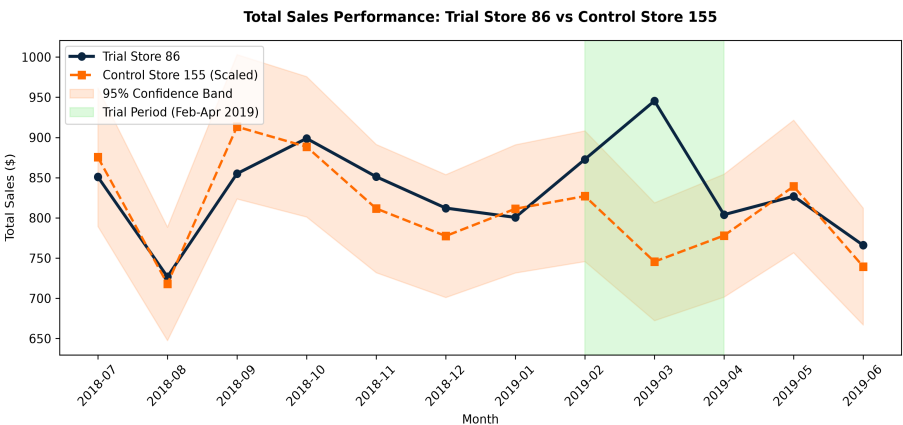
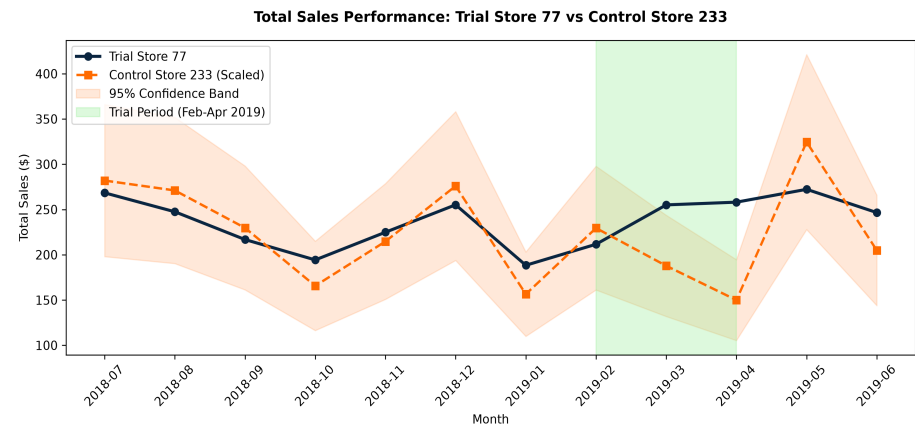
- **Welch's Hypothesis Test:** Mainstream Young Singles/Couples pay a significantly higher unit price (\$4.07/unit) compared to Budget & Premium peers (\$3.66/unit) — $t = 34.84$, $p < 10^{-200}$.
- **Brand Preference Vector:** High affinity for premium brands: Tyrrells (1.24x), Twisties (1.22x), Doritos (1.21x), Tostitos (1.21x), Kettle (1.19x), and Pringles (1.18x).
- **Pack Size Preference Vector:** Strong preference for sharing sizes: 270g (1.27x), 380g (1.26x), 330g (1.22x), and 210g (1.18x). Under-indexing on budget pack sizes.



3. Synthetic Control Matching & Trial Evaluation (Stores 77, 86, 88)

Difference-in-Differences (DiD) Causal Impact Analysis: Feb 2019 - Apr 2019

Trial Store	Matched Control	Pre-Trial Match Score	Sales Growth (%)	Sales t-stat (95% Sig)	Customer Growth (%)	Cust t-stat (95% Sig)
Store 77	Store 233	98.5%	Mar: +35.9% Apr: +72.1%	t = 5.93 (Yes)	Mar: +28.6% Apr: +70.4%	t = 15.14 (Yes)
Store 86	Store 155	92.5%	Mar: +26.8%	t = 6.68 (Yes)	Feb: +13.8% Mar: +18.3%	t = 5.37 (Yes)
Store 88	Store 237	76.7%	Mar: +25.7%	t = 3.38 (Yes)	Mar: +15.2%	t = 5.36 (Yes)



4. Strategic Recommendations & Rollout Roadmap

Actionable Strategy for Category Manager Julia:

1. FULL SYSTEM-WIDE ROLLOUT OF TRIAL LAYOUT:

- **Action:** Expand the experimental trial store shelf layout across all store locations in the network.
- **Financial Rationale:** Empirically proven to drive statistically significant sales increases of up to **+72.1%**, driven by higher customer foot traffic.

2. OPTIMIZE ASSORTMENT FOR MAINSTREAM YOUNG SINGLES/COUPLES:

- **Action:** Allocate premium shelf space to Kettle, Doritos, Pringles, Tyrrells, and 270g+ sharing sizes.
- **Placement Strategy:** Feature in high-visibility front-of-store endcaps and near beverage displays.

3. PROTECT VOLUME DRIVERS FOR OLDER FAMILIES (BUDGET):

- **Action:** Maintain family multi-packs to preserve high volume purchasing (~9.08 units/customer).